$\boxed{\Gamma \vdash T \cong T'}$ Types T and T' are convertible in context Γ

$$\textsc{ReflTy}\ \frac{\Gamma \vdash A}{\Gamma \vdash A \cong A} \qquad \textsc{SymTy}\ \frac{\Gamma \vdash A \cong B}{\Gamma \vdash B \cong A} \qquad \textsc{TransTy}\ \frac{\Gamma \vdash A \cong B \qquad \Gamma \vdash B \cong C}{\Gamma \vdash A \cong C}$$

$$\textsc{ElC}\ \frac{\Gamma \vdash A \cong A' : \mathcal{U}}{\Gamma \vdash A \cong A'} \qquad \textsc{FunTyC}\ \frac{\Gamma \vdash A \cong A' \qquad \Gamma, x{:}\, A \vdash B \cong B'}{\Gamma \vdash \Pi\, x{:}\, A.B \cong \Pi\, x{:}\, A'.B'}$$

$$\textsc{SigTyC}\ \frac{\Gamma \vdash A \cong A' \qquad \Gamma, x{:}\, A \vdash B \cong B'}{\Gamma \vdash \Sigma\, x{:}\, A.B \cong \Sigma\, x{:}\, A'.B'}$$

$$\textsc{IdTyC}\ \frac{\Gamma \vdash A \cong A' \qquad \Gamma \vdash t \cong t' : A \qquad \Gamma \vdash u \cong u' : A}{\Gamma \vdash \mathbb{I}\mathrm{d}(A, t, u) \cong \mathbb{I}\mathrm{d}(A', t', u')}$$

$\boxed{\Gamma \vdash t \cong t' : T}$ Terms t and t' are convertible at type T in context Γ

$$\textsc{Refl}\ \frac{\Gamma \vdash t : A}{\Gamma \vdash t \cong t : A} \qquad \textsc{Sym}\ \frac{\Gamma \vdash t \cong u : A}{\Gamma \vdash u \cong t : A} \qquad \textsc{Trans}\ \frac{\Gamma \vdash t \cong u : A \qquad \Gamma \vdash u \cong v : A}{\Gamma \vdash t \cong v : A}$$

$$\textsc{Conv}\ \frac{\Gamma \vdash t \cong t' : A \qquad \Gamma \vdash A \cong B}{\Gamma \vdash t \cong t' : B} \qquad \textsc{FunCong}\ \frac{\Gamma \vdash A \cong A' : \mathcal{U} \qquad \Gamma, x{:}\, A \vdash B \cong B' : \mathcal{U}}{\Gamma \vdash \Pi\, x{:}\, A.B \cong \Pi\, x{:}\, A'.B' : \mathcal{U}}$$

other congruences omitted

$$\beta\textsc{Fun}\ \frac{\begin{array}{c}\Gamma \vdash A \qquad \Gamma, x{:}\, A \vdash B \\ \Gamma, x{:}\, A \vdash t : B \qquad \Gamma \vdash u : A\end{array}}{\Gamma \vdash (\lambda\, x{:}\, A.t)\ u \cong t[u] : B[u]} \qquad \eta\textsc{Fun}\ \frac{\Gamma \vdash f : \Pi\, x{:}\, A.B}{\Gamma \vdash f \cong \lambda\, x{:}\, A.f\ x : \Pi\, x{:}\, A.B}$$

$$\beta\textsc{Sig}_1\ \frac{\begin{array}{c}\Gamma \vdash A \qquad \Gamma, x{:}\, A \vdash B \\ \Gamma \vdash t : A \qquad \Gamma \vdash u : B[t]\end{array}}{\Gamma \vdash \pi_1\ (t, u)_{A.B} \cong t : A} \qquad \beta\textsc{Sig}_2\ \frac{\begin{array}{c}\Gamma \vdash A \qquad \Gamma, x{:}\, A \vdash B \\ \Gamma \vdash t : A \qquad \Gamma \vdash u : B[t]\end{array}}{\Gamma \vdash \pi_2\ (t, u)_{A.B} \cong u : B[t]}$$

$$\eta\textsc{Sig}\ \frac{\Gamma \vdash A \qquad \Gamma, x{:}\, A \vdash B \qquad \Gamma \vdash p : \Sigma\, x{:}\, A.B}{\Gamma \vdash p \cong (\pi_1\ p, \pi_2\ p)_{A.B} : \Sigma\, x{:}\, A.B}$$

$$\beta\textsc{Zero}\ \frac{\Gamma, x{:}\, \mathbb{N} \vdash P \qquad \Gamma \vdash b_0 : P[0] \qquad \Gamma, x{:}\, \mathbb{N}, y{:}\, P[x] \vdash b_{\mathrm{S}} : P[\mathrm{S}(x)]}{\Gamma \vdash \mathrm{rec}_{\mathbb{N}}(0, x.P, b_0, x.y.b_{\mathrm{S}}) \cong b_0 : P[0]}$$

$$\beta\textsc{Succ}\ \frac{\Gamma \vdash n : \mathbb{N} \qquad \Gamma, x{:}\, \mathbb{N} \vdash P \qquad \Gamma \vdash b_0 : P[0] \qquad \Gamma, x{:}\, \mathbb{N}, y{:}\, P[x] \vdash b_{\mathrm{S}} : P[\mathrm{S}(x)]}{\Gamma \vdash \mathrm{rec}_{\mathbb{N}}(\mathrm{S}(n), x.P, b_0, x.y.b_{\mathrm{S}}) \cong b_{\mathrm{S}}[n, \mathrm{rec}_{\mathbb{N}}(n, x.P, b_0, x.y.b_{\mathrm{S}})] : P[\mathrm{S}(n)]}$$

$$\beta\textsc{Refl}\ \frac{\Gamma \vdash A \qquad \Gamma \vdash a : A \qquad \Gamma, x{:}\, A, y{:}\, \mathbb{I}\mathrm{d}(A, a, x) \vdash P \qquad \Gamma \vdash b : P[a, \mathrm{refl}_{A,a}]}{\Gamma \vdash \mathrm{rec}_{\mathbb{I}\mathrm{d}(A,a,\cdot)}(\mathrm{refl}_{A,a}, x.y.P, b) \cong b : P[a, \mathrm{refl}_{A,a}]}$$